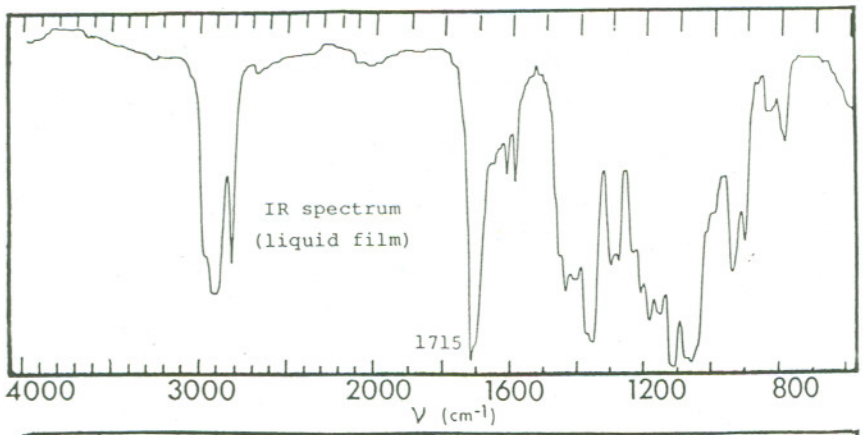
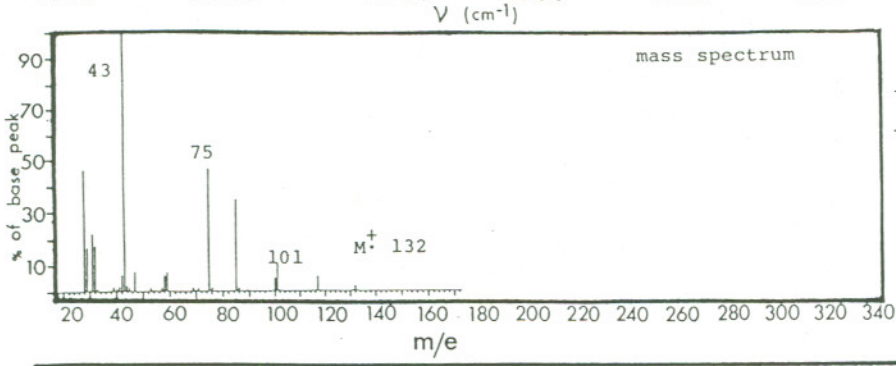
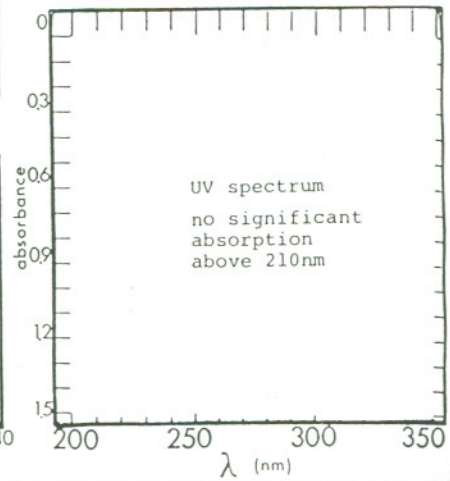
ORGANIC CHEMISTRY CUMULATIVE EXAM

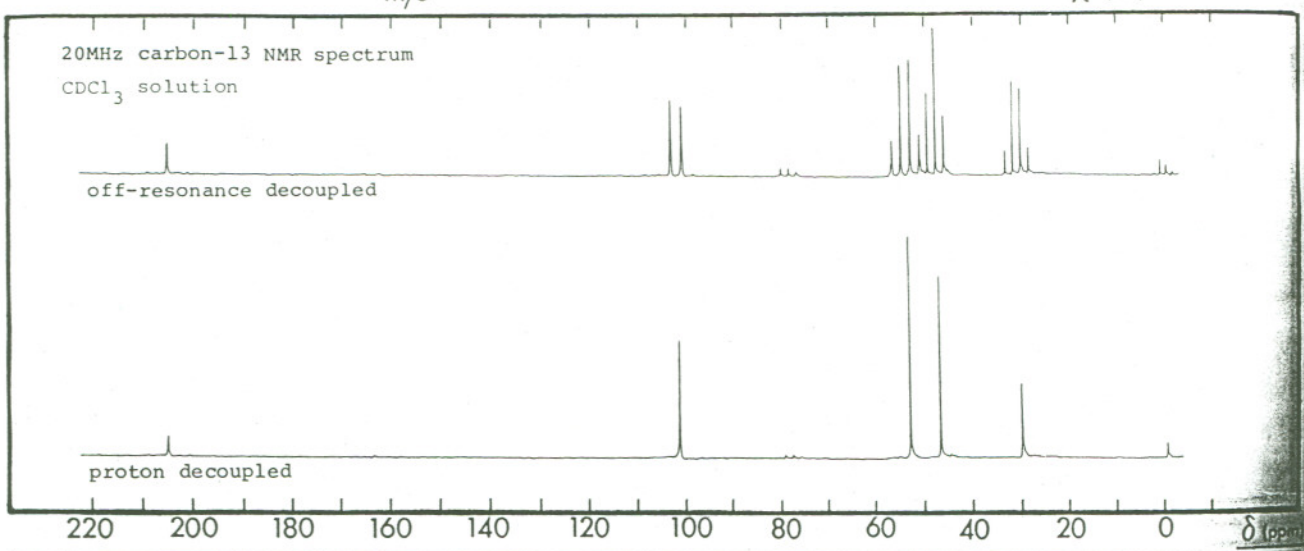
October 23, 1993

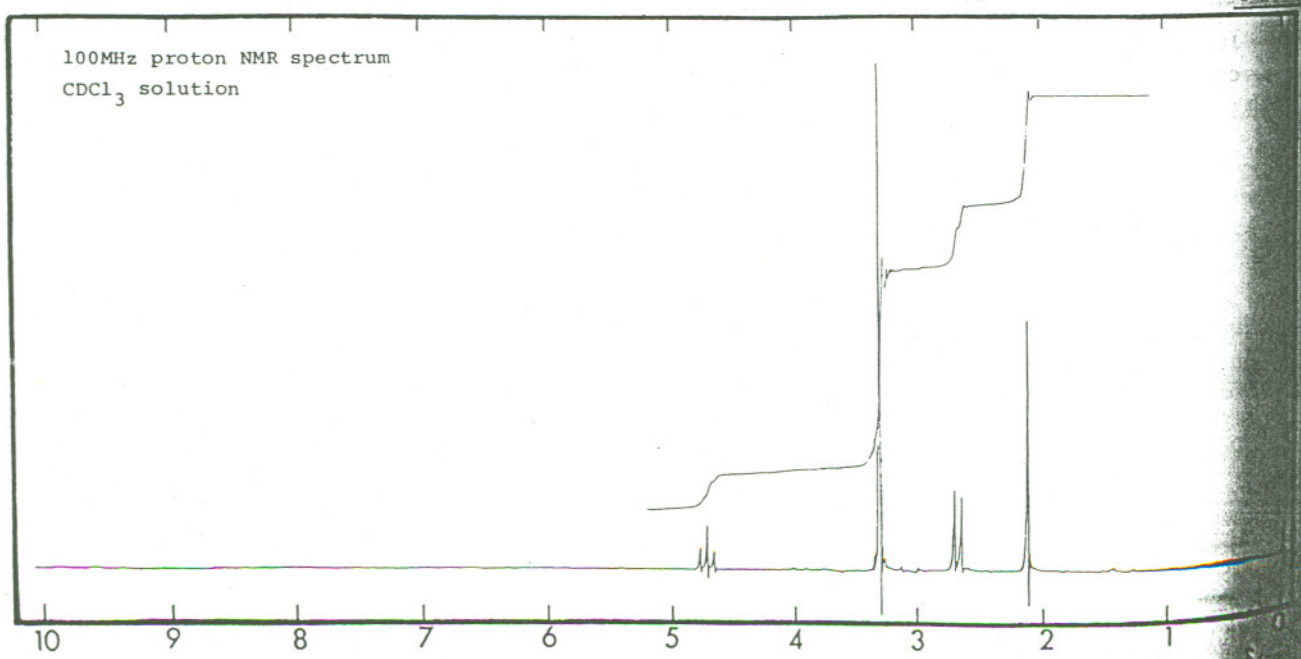
From the following three sets of organic spectra, provide complete chemical structures.

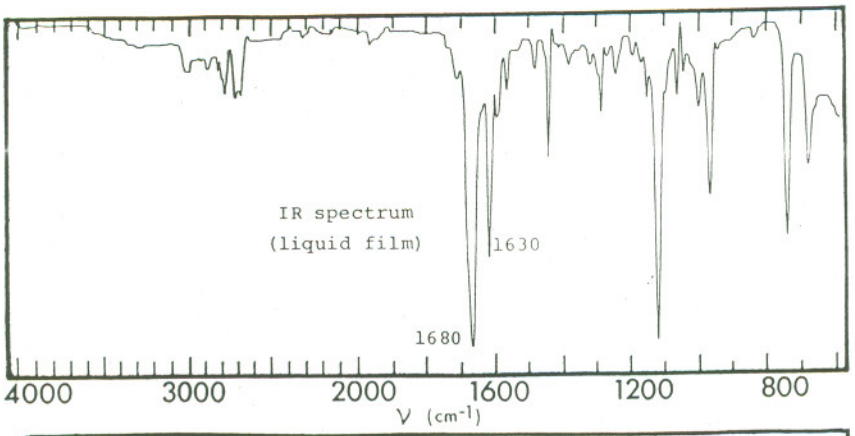

IR spectrum
(liquid film)
1715
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

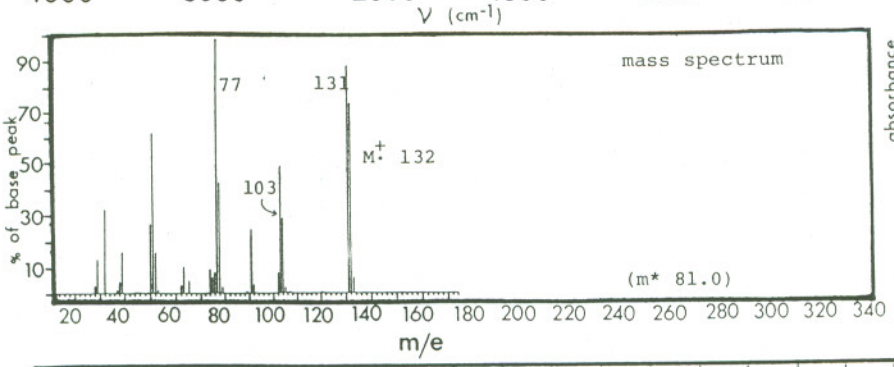

mass spectrum
43
75
101
M⁺ 132
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

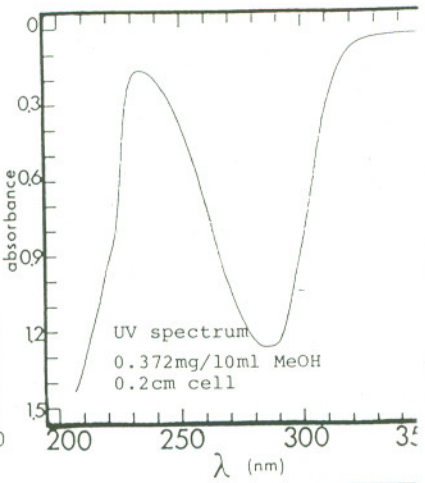

UV spectrum
no significant
absorption
above 210nm
absorbance
0
0.3
0.6
0.9
1.2
1.5
200
250
300
350
λ (nm)

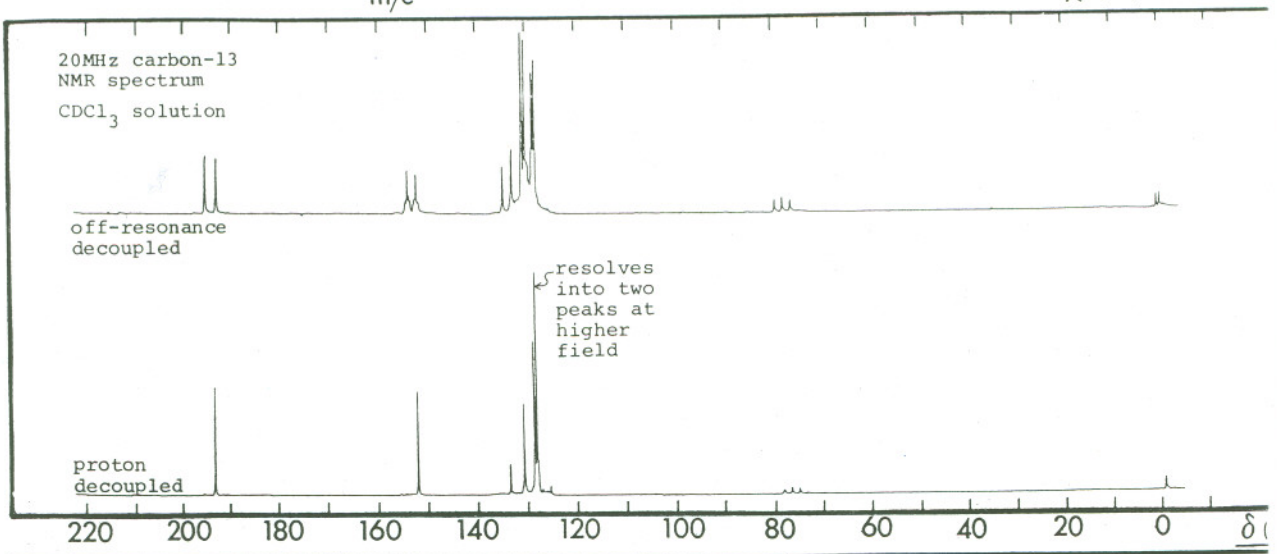

20MHz carbon-13 NMR spectrum
CDCl3 solution
off-resonance decoupled
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)

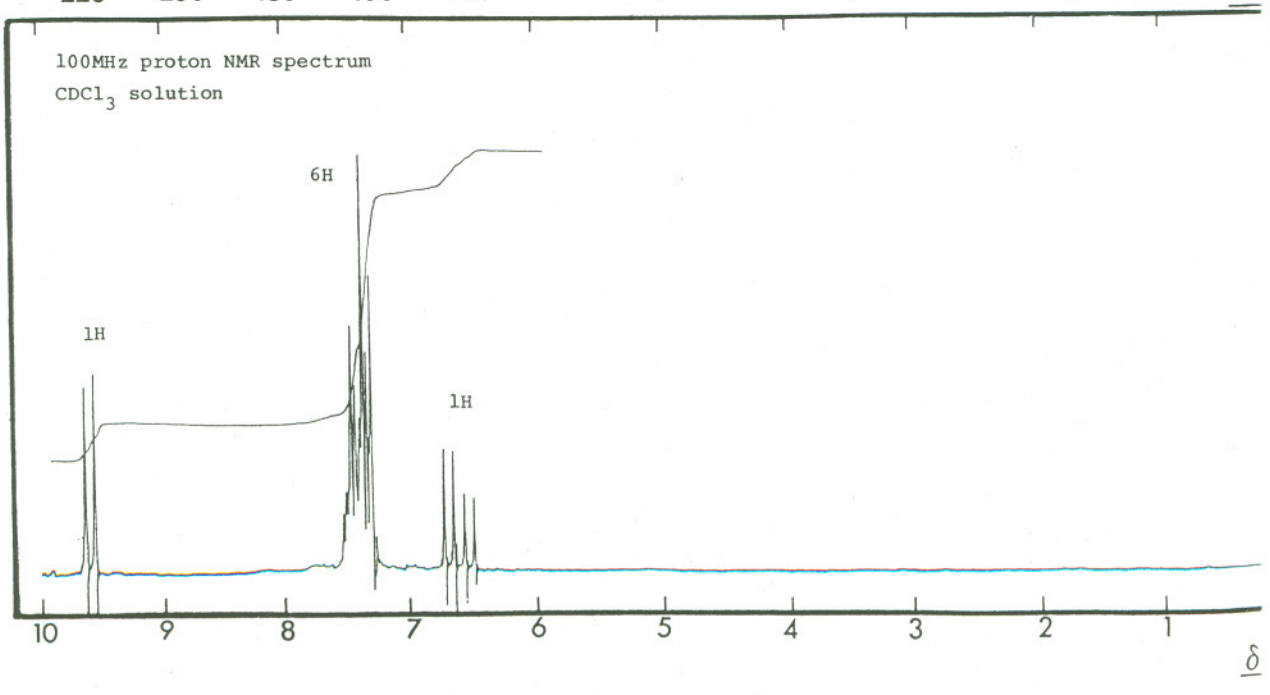

100MHz proton NMR spectrum
CDCl3 solution
10 9 8 7 6 5 4 3 2 1 0
δ

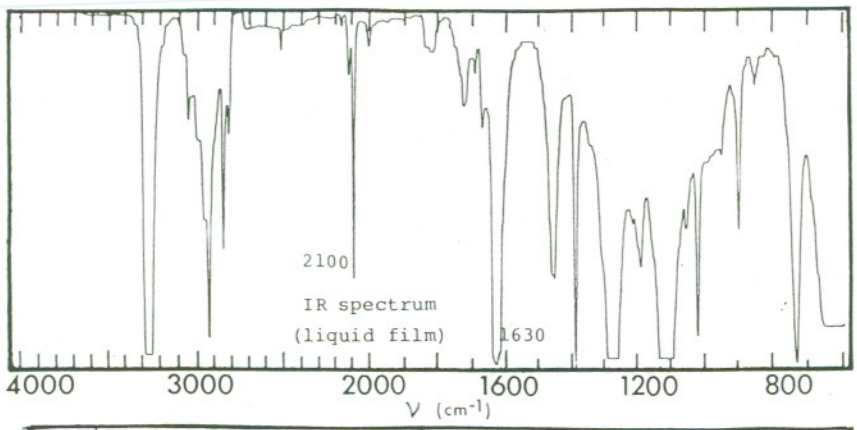

IR spectrum
(liquid film)
1630
1680
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

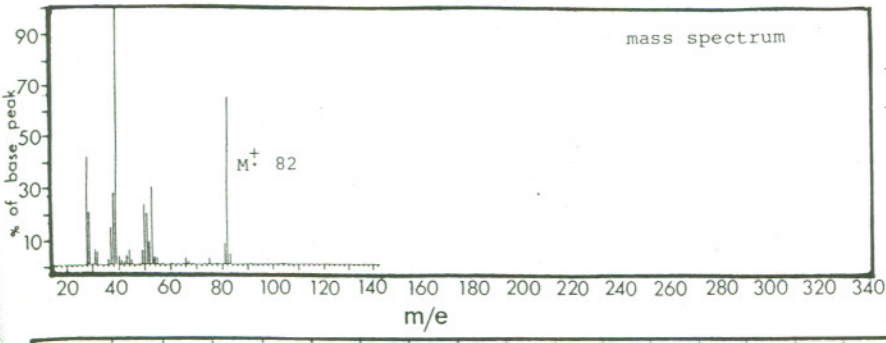

mass spectrum
77
131
M⁺ 132
103
(m* 81.0)
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

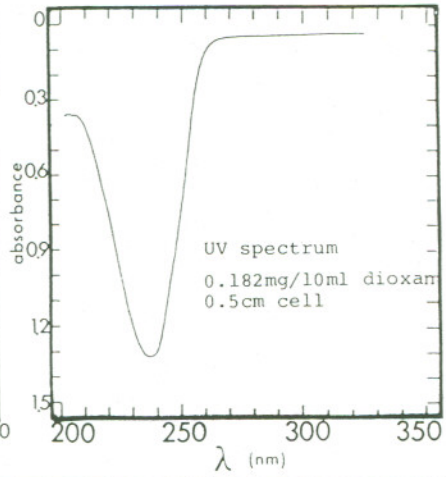

UV spectrum
0.372mg/10ml MeOH
0.2cm cell
absorbance
0
0.3
0.6
0.9
1.2
1.5
200
250
300
λ (nm)

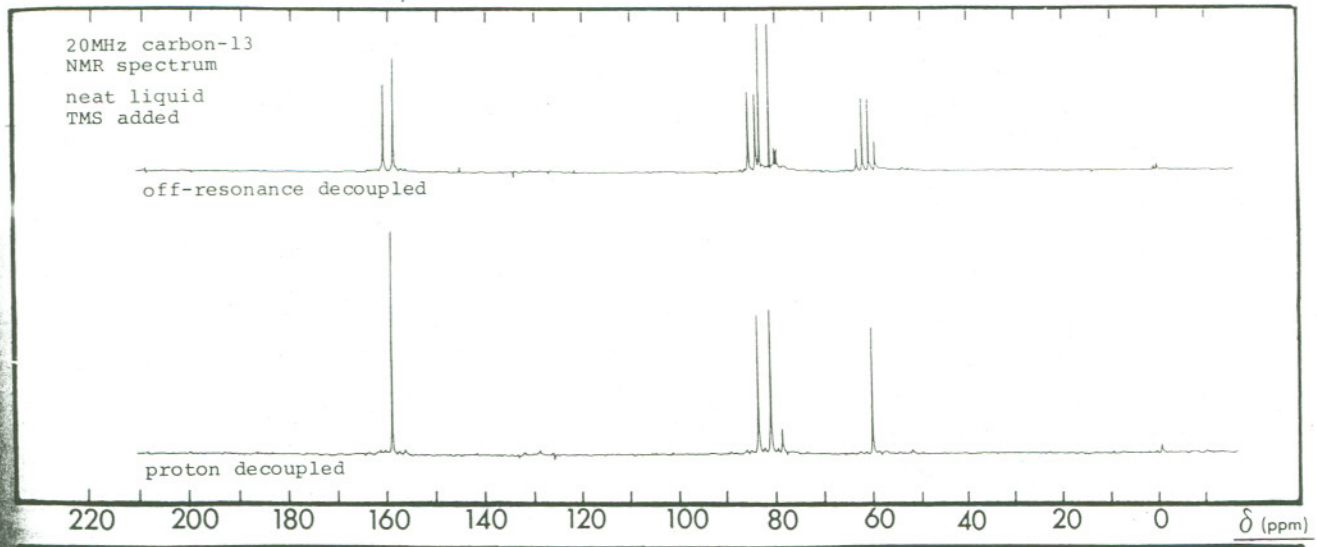

20MHz carbon-13
NMR spectrum
CDCl3 solution
off-resonance
decoupled
resolves
into two
peaks at
higher
field
proton
decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ

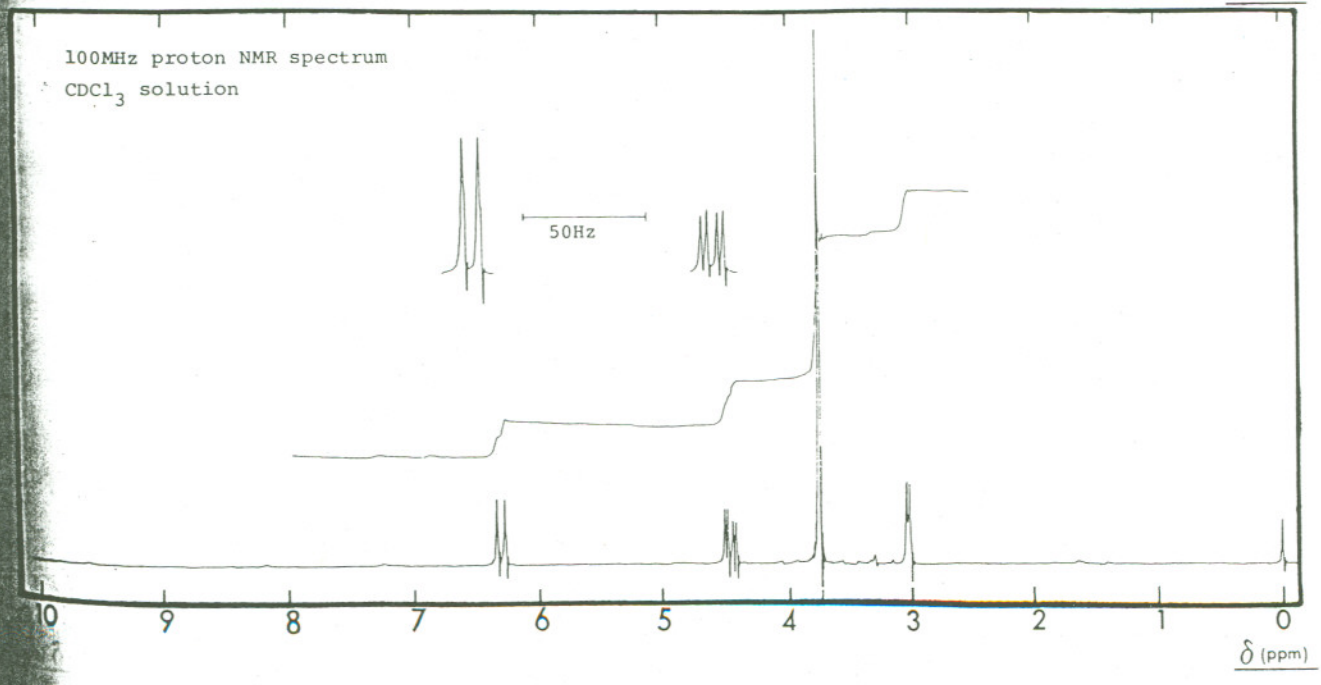

100MHz proton NMR spectrum
CDCl3 solution
6H
1H
1H
10 9 8 7 6 5 4 3 2 1
δ


2100
IR spectrum
(liquid film)
1630
4000
3000
2000
1600
1200
800
ν (cm⁻¹)


mass spectrum
M⁺· 82
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e


UV spectrum
0.182mg/10ml dioxan
0.5cm cell
absorbance
0
0.3
0.6
0.9
1.2
1.5
200
250
300
350
λ (nm)


20MHz carbon-13
NMR spectrum
neat liquid
TMS added
off-resonance decoupled
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)


100MHz proton NMR spectrum
CDCl3 solution
50Hz
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)